A Unified Sentinel-2 Imagery Thick Cloud Removal and Rescaling Framework From a Continuous Perspective

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***Abstract*—Cloud contamination presents a significant challenge in remote sensing of land surface, severely compromising data quality and hindering applications such as spectral unmixing, data fusion, and target detection. However, leveraging the full spectral bands of multitemporal Sentinel-2 images with different spatial resolutions (SRs) for thick cloud removal remains a longstanding challenge. To address this challenge, we propose a unified thick cloud removal and rescaling (UTCR2) framework from a continuous perspective, which can recover cloud-free multitemporal Sentinel-2 images with different SRs in a continuous manner. Specifically, we decompose the Sentinel-2 images with different SRs into the intrinsic abundance maps (i.e., the proportion of different materials at each pixel) with different SRs and discrete spectral signatures. To capture the intrinsic abundance maps with different SRs, we cleverly leverage the implicit neural representation (INR), which can naturally continuously represent data and fuse the spatial information across the full spectrum of multitemporal Sentinel-2 images with different SRs. To capture the discrete spectral signatures at different time nodes, we utilize fully connected networks (FCNs). Extensive simulated and real experiments on Sentinel-2 datasets demonstrate that the proposed UTCR2 significantly outperforms existing state-of-the-art methods and benefits downstream applications.**

***Index Terms*—Cloud removal, image reconstruction, implicit neural representation (INR), linear mixture model.**

I. Introduction

REMOTE sensing image is an indispensable tool for a broad spectrum of critical applications [1], [2], [3], including environmental monitoring [4], agriculture [5], disaster management [6], and urban planning [7]. However, the persistent challenge of cloud contamination severely limits the utility of such images. Clouds obscure essential features, impeding precise image analysis and the extraction of crucial information across various spectral bands. This issue is particularly detrimental in time-series analysis, where consistent, uninterrupted observations are crucial. For instance, accurately monitoring seasonal dynamics [8], land cover changes [9], and long-term environmental trends [10] requires a sequence of cloud-free images. Frequent interruptions in these datasets not only disrupt these analyses but also undermine the reliability of the conclusions drawn. Therefore, developing advanced and robust cloud-removal methods is essential to fully realize the potential of remote sensing technologies and enhance their applicability across these vital domains [11], [12].

In recent decades, significant advancements have been made in cloud removal techniques, which are essential for maximizing the utility of remote sensing data. These techniques primarily focus on reconstructing the information obscured by clouds, thereby reconstructing the integrity and usability of the imagery [13], [14]. For multitemporal Sentinel-2 imagery, cloud removal approaches are broadly categorized into two primary strategies: cloud removal on partial spectral bands with the same spatial resolution (SR) [15], [16] and cloud removal for full spectral bands after rescaling varying SRs to the same SR [17]. Cloud removal from partial spectral bands with the same SR typically involves stacking these bands to form a multispectral image (MSI), which is then processed using traditional thick cloud removal techniques [18], [19], [20]. By leveraging the spectral correlation within these stacked MSIs, numerous methods based on low-rank tensor representation have been developed to effectively address thick cloud contamination [16], [21]. However, since multitemporal Sentinel-2 imagery enjoys varying SRs, these methods exploit only partial spectral bands, thereby underutilizing the full spectral information and leading to unsatisfactory performance.

Cloud removal from full spectral bands after rescaling varying SRs to the same resolution first harmonizes all spectral bands to a unified SR using a rescaling algorithm [22], [23]. Then, these rescaled bands are stacked to form an MSI, which is subsequently processed using traditional thick cloud removal techniques [24], [25]. However, the effectiveness of these methods is constrained by the rescaling algorithms, which inevitably introduce spatial artifacts and thereby limit overall performance.

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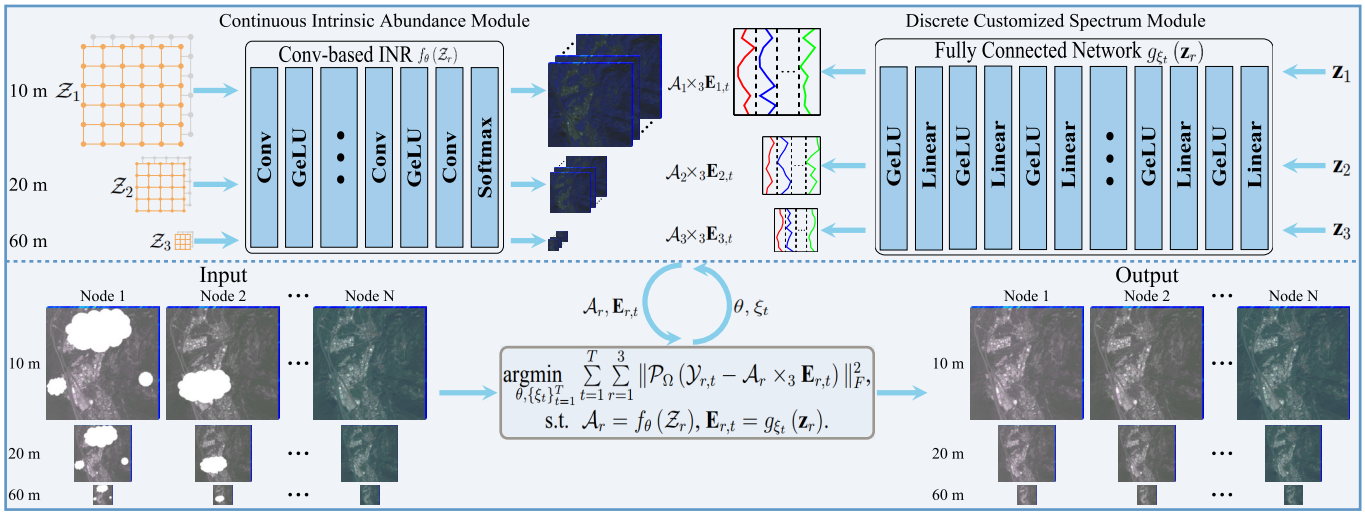


Fig. 1. Flowchart of the proposed UTCR2. The CIAM and DCSM are used to characterize the concatenated intrinsic abundance maps across various time nodes with different SRs (i.e., $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{A}_3$ which denote concatenated intrinsic abundance maps with 10, 20, and 60 m SR, respectively) and corresponding customized spectral signatures (i.e., $\{\mathbf{E}_{1,t}, \mathbf{E}_{2,t}, \mathbf{E}_{3,t}\}_{t=1}^T$).

A natural question is whether we can kill two birds with one stone, i.e., faithfully exploiting multitemporal Sentinel images with different SRs for thick cloud removal, which is a longstanding challenge. To answer this question, we suggest a unified thick cloud removal and rescaling (UTCR2) framework from a continuous perspective, allowing us to genuinely fuse full spectral bands of multitemporal Sentinel-2 images across different SRs and achieve promising performance. Specifically, inspired by the linear mixture model, we harness the power of implicit neural representations (INRs) [26], [27], which can essentially represent data across any arbitrary resolution [28], to capture the underlying abundance maps (e.g., the proportion of different materials [29]) of multitemporal Sentinel-2 images with different SRs. Meanwhile, we resort to dedicated fully connected networks (FCNs) to generate discrete spectral signals at various temporal nodes. By fusing multitemporal abundance maps with different SRs and corresponding spectral signatures, we reconstruct Sentinel-2 images across different spatial and temporal resolutions. An illustration of the proposed UTCR2 framework is provided in Fig. 1.

The main contributions of this work are as follows.

1) This article first suggests a unified framework for thick cloud removal and rescaling in Sentinel-2 imagery from a continuous perspective, effectively tackling the longstanding challenge of fully leveraging multitemporal Sentinel-2 images with varying SRs for cloud elimination.
2) To exploit multitemporal Sentinel-2 images with different SRs for cloud removal, we propose a UTCR2 framework from a continuous perspective. Specifically, we leverage the INR, which can continuously represent data and naturally fuse intrinsic abundance maps of multitemporal Sentinel-2 images with different SRs, to capture underlying abundance maps. Meanwhile, we utilize FCNs to capture corresponding discrete spectral signatures at different time nodes.
3) The proposed framework can simultaneously achieve cloud removal and rescaling. Comprehensive evaluations on both simulated and real-world datasets validate the effectiveness of the UTCR2 framework, demonstrating its superior performance over state-of-the-art cloud removal methods in both visual quality and numerical assessments.

II. Related Work

In the literature, many thick cloud removal methods for multitemporal Sentinel-2 images have been proposed, most of which focus on removing thick clouds from partial spectral bands with the same SR [18], [30]. For example, He et al. [31] stacked visible and near-infrared bands from time-series Sentinel-2 images to create a multitemporal MSI, subsequently applying a total variation-regularized low-rank tensor ring completion model for cloud removal. Similarly, Chen et al. [32] utilized a low-rank tensor representation to extract cloud-free MSIs, supplemented by a sparse representation to capture cloud components. Recently, the exceptional expressive power of untrained neural networks has garnered increasing interest, leading to the development of deep learning-based techniques for thick cloud removal [33]. For instance, Ji et al. [34] employed two separate fully convolutional neural networks (CNNs) to independently detect cloud masks and remove thick clouds. Jiang et al. [35] addressed the problem using a two-stage approach: the coarse stage applies frequency-domain processing and attention mechanisms for targeted feature fusion, while the refined stage employs a network that enhances reconstruction through image self-correlation. However, not all spectral bands are leveraged in these methods, resulting in unsatisfactory performance.

To exploit the full spectral bands of multitemporal Sentinel-2 images, many researchers employ thick cloud methods after rescaling varying SRs to the same SR using a rescaling algorithm [36], [37], [38]. Meraner et al. [39] developed a residual neural network to remove clouds from rescaled Sentinel-2 images. Their approach was further enhanced by incorporating synthetic aperture radar (SAR) imagery as auxiliary data, significantly improving cloud removal performance. Building

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on this foundation, Ebel et al. [40] employed a 3-D CNN to predict cloud-free images from sequences of cloudy optical and SAR data. To achieve sharper image reconstructions, they introduced a perceptual loss function [41] during the training of their 3-D CNN. However, acquiring both rescaled optical and SAR imagery for the same region presents significant challenges. To address this limitation, Wang et al. [25] proposed an iterative diffusion process that requires only rescaled optical imagery for cloud removal. Similarly, Zou et al. [24] utilized conditional guided diffusion combined with deep convolutional networks to achieve high-performance cloud removal in optical Sentinel-2 imagery.

III. Preliminaries

In this article, we adopt the following notations: scalars, vectors, matrices, and tensors are denoted by lowercase letters (e.g., x), bold lowercase letters (e.g., $\mathbf{x}$), bold uppercase letters (e.g., $\mathbf{X}$), and calligraphic uppercase letters (e.g., $\mathcal{X}$), respectively. The $(i_1, i_2, \ldots, i_N)$th element of an Nth order tensor $\mathcal{X}$ is represented as $\mathcal{X}(i_1, i_2, \ldots, i_N)$. The Hadamard product of two tensors $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ and $\mathcal{B} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is denoted by $\mathcal{C} = \mathcal{A} \odot \mathcal{B}$, where $\mathcal{C}(i_1, i_2, \ldots, i_N) = \mathcal{A}(i_1, i_2, \ldots, i_N)\mathcal{B}(i_1, i_2, \ldots, i_N)$. The Frobenius norm of a tensor $\mathcal{A}$ is defined as $\|\mathcal{A}\|_F = \left(\sum_{i_1, i_2, \ldots, i_N} \mathcal{A}(i_1, i_2, \ldots, i_N)^2\right)^{1/2}$.

Definition 1 (Mode-n Product [42]) The mode-n tensor-matrix product of an Nth order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ and a matrix $\mathbf{B} \in \mathbb{R}^{J \times I_n}$ is denoted by $\mathcal{C} = \mathcal{X} \times_n \mathbf{B}$, where $\mathcal{C} \in \mathbb{R}^{I_1 \times \cdots \times I_{n-1} \times J \times I_{n+1} \times \cdots \times I_N}$. This operation involves multiplying the mode-n unfolding of $\mathcal{X}$ by $\mathbf{B}$, such that the $(i_1, i_2, \ldots, i_N)$th element of $\mathcal{X}$ is appropriately mapped to the corresponding element in the resulting tensor $\mathcal{C}$.

IV. Proposed Method

We first introduce the overarching framework for thick cloud removal in multitemporal remote sensing imagery. Then, we present the proposed UTCR2, offering a detailed exploration of its key components, with particular emphasis on the intricately designed continuous intrinsic abundance module (CIAM) and discrete customized spectrum module (DCSM). Finally, we outline the algorithm and provide the pseudocode used to solve the UTCR2-based cloud removal problem.

A. Problem Formulation

Assuming that clean Sentinel-2 images are obscured by thick clouds, the degradation model can be expressed as

$$\mathcal{Y} = \mathcal{P}_\Omega(\mathcal{X})$$

where $\mathcal{Y}$ represents the cloud-contaminated observation, and $\mathcal{X}$ denotes the underlying cloud-free image. The operator $\mathcal{P}_\Omega$ is a projection operator that preserves pixel values at the cloud-free locations specified by Ω, while setting the cloud-covered pixels to zero. Here, Ω corresponds to the set of coordinates indicating cloud-free areas. The goal of thick cloud removal is to reconstruct the original cloud-free image from the cloud-contaminated observation. This problem is inherently ill-posed, requiring the incorporation of suitable priors for effective solution. Mathematically, it is commonly formulated as

$$\min_{\mathcal{X}} \frac{1}{2}\|\mathcal{P}_\Omega(\mathcal{X} - \mathcal{Y})\|_F^2 + \lambda \mathcal{R}(\mathcal{X})$$

where $\mathcal{R}(\mathcal{X})$ is the regularization term that encapsulates prior knowledge about the Sentinel-2 image, and λ is the associated regularization parameter.

Inspired by the linear mixture model, a single-temporal remote sensing image can be decomposed into abundance maps and their corresponding spectral signatures. Mathematically, this decomposition is represented as the mode-3 product of the abundance maps and the corresponding spectral signatures. Thus, the Sentinel-2 imagery at each tth time node can be expressed as

$$\mathcal{X}_t = \mathcal{A}_t \times_3 \mathbf{E}_t$$

where $\mathcal{A}_t$ represents the intrinsic abundance maps of $\mathcal{X}_t$, and $\mathbf{E}_t$ represents the spectral signatures of $\mathcal{X}_t$. Here, $\times_3$ denotes the mode-3 product.

Unfortunately, different spectral bands of multitemporal Sentinel-2 images feature different SRs. How to faithfully exploit multitemporal Sentinel-2 images across different SRs for thick cloud removal remains a difficult challenge.

B. Overview of the Proposed UTCR2

To meet this challenge, we propose a unified continuous framework (UTCR2), which allows us to faithfully exploit multitemporal Sentinel-2 images across different SRs. Specifically, the proposed UTCR2 is composed of two key modules, i.e., CIAM and DCSM. The CIAM, which is essentially a continuous function, maps meshed spatial coordinates to abundances. By feeding meshed spatial coordinates of any arbitrary resolution into the continuous abundance function, we can obtain the abundance maps with corresponding SRs, including 10, 20, and 60 m SRs. The DCSM maps the spectral coordinates and timestamp t to the spectral signatures at the tth time node. By integrating the abundance maps with different SRs and corresponding spectral signatures, we can obtain multitemporal Sentinel-2 images with different SRs.

Mathematically, given an observed Sentinel-2 image $\mathcal{Y}$, the formulation of the proposed UTCR2-based model for thick cloud removal is as follows:

$$\begin{aligned} &\arg\min_{\theta, \{\xi_t\}_{t=1}^T} \sum_{h,w,b,t} \|\mathcal{P}_\Omega\left(\mathcal{Y}(h,w,b,t) - \mathcal{X}(h,w,b,t)\right)\|_F^2 \\ &\text{s.t. } \mathcal{X}(h,w,b,t) = \sum_{s=1}^{S} \mathcal{A}(h,w,s,t) \cdot \mathbf{E}_t(b,s) \\ &\mathcal{A}(h,w,:) = f_\theta(h,w); \mathbf{E}_t(b,:) = g_{\xi_t}(b) \end{aligned} \tag{1}$$

where $\mathcal{Y}(h,w,b,t)$ and $\mathcal{X}(h,w,b,t)$ denote the (h,w,b,t)-th elements of observed Sentinel-2 image $\mathcal{Y}$ and reconstructed Sentinel-2 image $\mathcal{X}$, respectively; (h,w) denotes the spatial coordinates; b denotes the spectral coordinates; t denotes the timestamps; The functions $f_\theta(\cdot)$ denote the CIAM, which maps the spatial coordinate (h,w) to the intrinsic abundance $\mathcal{A}(h,w,:)$ at corresponding position; $g_{\xi_t}(\cdot)$ denotes the tth DCSM, which maps the spectral coordinate b to the spectral

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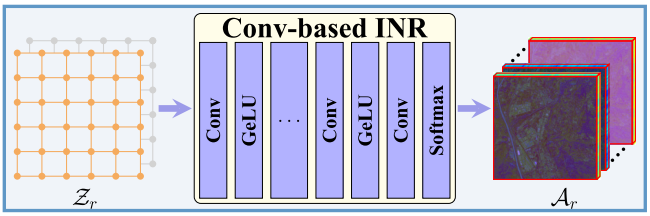


Fig. 2. Detailed architecture of the CIAM.

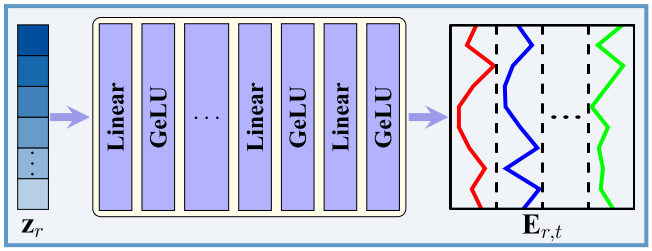


Fig. 3. Detailed architecture of each DCSM.

signatures at tth time node. Finally, θ and $\{\xi_t\}_{t=1}^T$ are the learnable parameters of the CIAM and DCSM, respectively.

For simplicity, we rewrite the problem (1) as follows:

$$\begin{aligned} &\arg\min_{\theta,\{\xi_t\}_{t=1}^T} \sum_{t=1}^{T}\sum_{r=1}^{3} \left\| \mathcal{P}_\Omega\left(\mathcal{Y}_{r,t} - \mathcal{A}_r \times_3 \mathbf{E}_{r,t}\right)\right\|_F^2 \\ &\text{s.t. } \mathcal{A}_r = f_\theta(\mathcal{Z}_r); \quad \mathbf{E}_{r,t} = g_{\xi_t}(\mathbf{z}_r). \end{aligned} \tag{2}$$

Here, $\mathcal{Y}_{r,t}$ for $r = 1,2,3$ denote the cloud-contaminated images with 10, 20, and 60 m SRs at the tth time node, respectively. The tensor $\mathcal{A}_r$ for $r = 1,2,3$ represents the abundance map with 10, 20, and 60 m SRs, respectively. The vector $\mathbf{E}_{r,t}$ represents the spectral signatures at the tth time node. $\mathcal{Z}_r \in \mathbb{R}^{H_1 \times W_1 \times 2}$ and $\mathbf{z}_r \in \mathbb{R}^{B_r \times 1}$ are the meshed spatial coordinates and vectorized spectral coordinates, respectively.

The flowchart of the proposed UTCR2 framework for multitemporal Sentinel-2 images is illustrated in Fig. 1. In the following sections, we provide a detailed explanation of the key components of the UTCR2 framework, specifically focusing on the CIAM $f_\theta(\cdot)$ and the DCSM $g_{\xi_t}(\cdot)$.

1) Continuous Intrinsic Abundance Module: To seamlessly exploit multitemporal Sentinel-2 images across different SRs, we leverage INR to design a CIAM $f_\theta(\cdot)$ with learnable parameters θ. The CIAM inherits the fundamental properties of INR, enabling the natural representation of data at arbitrary resolutions to characterize the abundance maps underlying Sentinel-2 images with varying SRs without relying on any rescaling algorithm. Building upon the continuous nature of INR, the CIAM is designed to map meshed spatial coordinates directly to abundances, allowing for seamless integration across different SRs. The detailed structure of the CIAM is illustrated in Fig. 2.

Specifically, the CIAM consists of several convolutional layers with nonlinear activation functions, and concludes with a fully connected layer. The input to the CIAM is the meshed spatial coordinates $\mathcal{Z}_r$, which enjoys the same SR as the target abundances $\mathcal{A}_r$. Each pixel coordinate consists of a vertical and a horizontal coordinate, which are both normalized to the range $[-1, 1]$. For an image of the size $H_r \times W_r$, we can obtain the meshgrid spatial coordinates of the size $H_r \times W_r \times 2$. By feeding the corresponding spatial coordinates $\mathcal{Z}_r$ into the module, we can obtain the intrinsic abundances $\mathcal{A}_r \in \mathbb{R}^{H_r \times W_r \times R \times T}$ with the corresponding SR. Mathematically, $\mathcal{A}_r$ can be formulated as follows:

$$\mathcal{A}_r = f_\theta(\mathcal{Z}_r) = \sigma^L \circ \phi^L \circ \cdots \circ \sigma^2 \circ \phi^2 \circ \sigma^1 \circ \phi^1(\mathcal{Z}_r) \tag{3}$$

where $\sigma^l(\cdot)$ and $\phi^l(\cdot)$ denote the nonlinear activation function and convolution layer for $l = 1,2,\ldots,L$, respectively; L is the number of convolution layers. For $l = 1,2,\ldots,L-1$, $\sigma^l(\cdot)$ denotes commonly used activation function GeLU. $\sigma^L(\cdot)$ is Softmax function which is used to normalize the abundances $\mathcal{A}_r$.

Different from FCN-based INRs using fully connected layers, convolution-based INR uses convolution layers as the backbone, conducting convolution with a kernel size of $k \times k$ on the corresponding input meshed spatial coordinates [43]. Especially, when the kernel size is set to be $k = 1$, convolution-based INR can degrade to FCN-based INR. Convolution-based INR considers adjacent pixels together instead of in isolation, which can effectively model and leverage the locality prior. Hence, compared with FCN-based INR, convolution-based INR can capture more accurate abundance maps, leading to higher-quality reconstruction of multitemporal Sentinel-2 images.

2) Discrete Customized Spectrum Module: To capture the customized spectral signatures at tth time node, we leverage a FCN to elaborately design the DCSM $g_{\xi_t}(\cdot)$ with learnable parameters ξ_t. Empowered with the FCN, the DCSM can capture the complex spectral signatures and faithfully utilize full spectral bands to reconstruct multitemporal Sentinel-2 images, leading to promising reconstruction performance. The detailed structure of the DCSM is illustrated in Fig. 3.

Specifically, we elaborately design T DCSMs to capture the spectral signatures at T timestamps. Each DCSM is composed of several fully connected layers activated by a nonlinear activation function. The tth DCSM receives a vectorized spectral coordinates $\mathbf{z}_r \in \mathbb{R}^{B_r \times 1}$ as input and outputs the spectral signatures at tth time node, i.e., $\mathbf{E}_{r,t} \in \mathbb{R}^{B_r \times R}$, B_r means the total number of the spectral bands at rth resolution. The mathematical formulation of $\mathbf{E}_{r,t}$ is

$$\mathbf{E}_{r,t} = g_{\xi_t}(\mathbf{z}_r) = \sigma^L \circ \psi^L \circ \cdots \circ \sigma^2 \circ \psi^2 \circ \sigma^1 \circ \psi^1(\mathbf{z}_r) \tag{4}$$

where $\sigma^l(\cdot)$ and $\psi^l(\cdot)$ denote the nonlinear activation function and fully connected layer for $l = 1,2,\ldots,L$, respectively; L is the number of fully connected layers.

C. Solving Algorithm

To tackle the highly nonlinear and nonconvex problem (1), we employ the adaptive moment estimation (Adam) optimizer [44]. The optimization process is guided by a predefined maximum number of iterations, $k_{\max}$, which serves as the stopping criterion. Pseudocode for the Adam-based algorithm used to solve the proposed UTCR2-based model is summarized in Algorithm 1.

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Algorithm 1 UTCR2 for Thick Cloud Removal

Input: Observed image $\mathcal{Y}$, Ω, number of endmembers N.
Output: Cloud-free $\mathcal{X}$;
For $k = 1, 2, \ldots, k_{max}$
    $\mathcal{A}_r = f_{\theta^{k-1}}(\mathcal{Z}_r), \mathbf{E}_{r,t} = g_{\xi_t^{k-1}}(\mathbf{z}_r)$ for $r = 1, 2, 3$ and $t = 1, 2, \ldots, T$;
    Update θ and $\{\xi_t\}_{t=1}^{T}$ via Adam;
End for
return $\mathcal{X}_{r,t} = \mathcal{P}_{\Omega}(\mathcal{Y}_{r,t}) + \mathcal{P}_{\overline{\Omega}}((\mathcal{A}_r \times_3 \mathbf{E}_{r,t}))$ for $r = 1, 2, 3$, and $t = 1, 2, \ldots, T$.

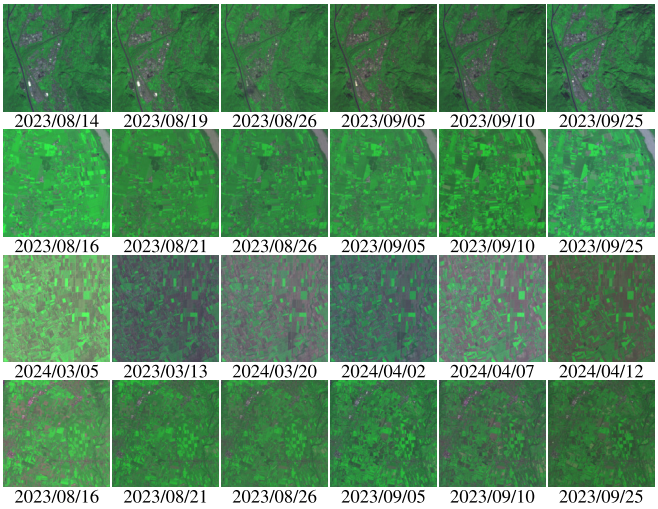


Fig. 4. False-color images composed of bands 3, 4, and 2 of Sentinel-2 images for simulated experiments. (Top to bottom) *T32TPS*, *T32TQS*, *T33TUL*, and *T33TUM*, respectively.

V. Experiments

A. Experimental Setting

To evaluate the performance of the proposed UTCR2, we conduct comprehensive experiments using both simulated and real-world multitemporal remote sensing images. Our method is compared with three classical low-rank tensor modeling-based cloud removal methods (i.e., HaLRTC [45], TVLRSD [32], and RTCR [46]), a deep learning-based method (i.e., STLR-DP [47]), and an interpolation-based method (i.e., VPint [18]). All the competing methods are proposed for thick cloud removal from partial spectral bands with the same SR. For all competing methods, we manually adjust the parameters to achieve optimal performance based on the authors' recommendations. Experiments are performed on a system running Windows 11, equipped with an 11th Gen Intel Core i9-9900KF at 3.60 GHz CPU, an NVIDIA GeForce RTX 2080Ti GPU, and 64 GB of RAM. The implementations of STLR-DP and UTCR2 are conducted using the PyTorch platform with Python, while HaLRTC, TVLRSD, and RTCR are implemented using MATLAB.

In our simulated experiments, we first download four cloud-free Sentinel-2 images from the website.[1] Each image comprises 13 spectral bands at each time node. Specifically, the spectral bands exhibit three different SRs: 10 m (B2, B3, B4, B8), 20 m (B5, B6, B7, B8A, B11, B12), and 60 m (B1, B9, B10). Then, we crop the subimage of the spatial size 600×600, 300×300, and 100×100 at 10, 20, and 60 m SRs, respectively. Then, we manually generate the cloud mask at 10 m SR. The cloud masks at 20 and 60 m SRs are obtained by nearest downsampling the cloud mask at 10 m SR. For each Sentinel-2 image, three temporal images were selected to simulate cloud cover, serving as cloud-contaminated images, while the remaining three images were used as reference images. All pixel values were normalized to the [0, 1] range band-by-band. The experiments were conducted using Sentinel-2 imagery from six different time nodes. Pseudocolor images of the datasets are presented in Fig. 4.

[1] https://browser.dataspace.copernicus.eu/

B. Simulated Experiments

To evaluate the performance of different cloud removal methods, we employed five commonly used quantitative metrics: peak signal-to-noise ratio (PSNR) [48], structural similarity index (SSIM) [48], universal image quality index (UIQI) [49], correlation coefficient (CC) [50], and root mean square error (RMSE). Typically, higher values of MPSNR, MSSIM, UIQI, and CC signify better restoration quality, as they indicate closer alignment with the original image. Conversely, a lower RMSE value suggests a smaller deviation between the restored and original images, reflecting higher accuracy in the reconstruction process.

Table I displays the PSNR, SSIM, UIQI, CC, and RMSE metrics for reconstruction results achieved by various methods on Sentinel-2 datasets. From Table I, we can observe significant improvements in terms of PSNR values for all methods compared with the observed images. HaLRTC and TVLRSD show similar performance in most scenarios, while RTCR frequently surpasses both, achieving considerable improvements. STLR-DP consistently secures the second position in almost all simulated data evaluations, benefiting from the strong representational capacity of deep neural networks. Furthermore, the proposed UTCR2 consistently surpasses all other methods across every evaluation metric, which can be attributed to its flexible representation crossing different SRs and effective utilization of the full spectral information.

Fig. 5 presents the visual reconstruction results from various methods, accompanied by the corresponding reconstruction results and original images from the simulated experiments. For better visualization, a highlighted region of interest, marked with a red square, is enlarged in the lower-right section of every Sentinel-2 image. As shown in Fig. 5, HaLRTC successfully reconstructs the main structures of the cloud-affected areas but struggles to recover finer spatial details. TVLRSD, on the other hand, captures more spatial details than HaLRTC, though its results show noticeable discrepancies in both geometric structure and color fidelity compared to the original images. RTCR delivers visually appealing results in cloud-contaminated regions except *T33TUM*. STLR-DP blurs some local details and shows some noticeable discrepancies in color fidelity. In contrast, the proposed UTCR2, which utilizes a flexible representation to model abundance maps of Sentinel-2 images at any SR and fully exploits spectral bands, multitemporal resolutions, and the inherent physical

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TABLE I

AVERAGE NUMERICAL RESULTS OF THICK CLOUD REMOVAL BY DIFFERENT METHODS ON T32TPS, T32TQS, T33TUL, AND T33TUM. THE BEST AND SECOND-BEST VALUES ARE HIGHLIGHTED

Time Node		Node 1					Node 2					Node 3				
SR	Methods	PSNR	SSIM	UIQI	CC	RMSE	PSNR	SSIM	UIQI	CC	RMSE	PSNR	SSIM	UIQI	CC	RMSE
10 m	Observed	18.580	0.1546	0.0204	0.076	0.1244	20.723	0.1702	0.0360	0.1443	0.0961	18.823	0.2291	0.0087	0.0201	0.1267
	HaLRTC	46.080	0.9875	0.9603	0.9613	0.0056	40.043	0.9796	0.7293	0.7713	0.0187	40.420	0.9659	0.7426	0.7724	0.0175
	TVLRSD	45.094	0.9902	0.9462	0.9531	0.0064	41.561	0.9940	0.8828	0.9135	0.0097	44.718	0.9739	0.8985	0.9180	0.0091
	RTCR	48.850	0.9944	0.9776	0.9784	0.0042	44.592	0.9930	0.8291	0.8591	0.0104	44.631	0.9798	0.8299	0.8533	0.0164
	STLR-DP	47.619	0.9954	0.9645	0.9676	0.0051	51.267	0.9985	0.9704	0.9771	0.0041	51.519	0.9923	0.9829	0.9830	0.0037
	VPint	40.539	0.9850	0.8422	0.8777	0.0107	41.199	0.9889	0.8478	0.8800	0.0111	40.744	0.9859	0.8043	0.8468	0.0099
	UTCR2	**52.789**	**0.9969**	**0.9916**	**0.9916**	**0.0026**	**54.867**	**0.9989**	**0.9858**	**0.9887**	**0.0029**	**54.986**	**0.9947**	**0.9895**	**0.9895**	**0.0029**
20 m	Observed	13.827	0.1445	0.0232	0.0861	0.2242	17.287	0.1613	0.0403	0.1602	0.1406	16.116	0.2205	0.0166	0.0475	0.1694
	HaLRTC	39.511	0.9641	0.9568	0.9580	0.0113	36.093	0.9605	0.7584	0.7952	0.0215	36.918	0.9383	0.7858	0.8145	0.0211
	TVLRSD	36.462	0.9347	0.8500	0.8991	0.0227	36.565	0.9714	0.8435	0.8885	0.0152	40.421	0.9699	0.9334	0.9384	0.0151
	RTCR	38.632	0.9751	0.9231	0.9255	0.0155	40.872	0.9847	0.8915	0.9067	0.0115	40.496	0.9753	0.9037	0.9116	0.0420
	STLR-DP	41.805	0.9874	0.9617	0.9641	0.0091	48.507	0.9969	0.9802	0.9823	0.0049	45.220	0.9862	0.9782	0.9782	0.0070
	VPint	37.796	0.9853	0.9470	0.9527	0.0146	40.210	0.9810	0.9389	0.9473	0.0126	40.841	0.9832	0.9337	0.9420	0.0104
	UTCR2	**46.482**	**0.9913**	**0.9904**	**0.9904**	**0.0052**	**51.062**	**0.9977**	**0.9867**	**0.9873**	**0.0041**	**49.524**	**0.9894**	**0.9882**	**0.9882**	**0.0051**
60 m	Observed	4.628	0.1102	0.0035	0.0238	0.6358	8.017	0.1345	0.0155	0.0821	0.5033	5.039	0.1823	0. 0775	0.0225	0.5949
	HaLRTC	45.306	0.9755	0.9369	0.9399	0.0073	41.191	0.9646	0.6213	0.7016	0.0136	42.542	0.9398	0.8036	0.8234	0.0142
	TVLRSD	38.881	0.9592	0.7363	0.7634	0.0154	39.070	0.9775	0.5081	0.5861	0.0165	38.861	0.9376	0.7148	0.7903	0.0176
	RTCR	42.655	0.9805	0.7974	0.8329	0.0087	35.269	0.9511	0.5262	0.5999	0.0319	37.559	0.9506	0.7336	0.7493	0.0262
	STLR-DP	41.782	0.9656	0.8102	0.8273	0.0253	39.425	0.9771	0.5331	0.6002	0.0176	35.808	0.9487	0.7195	0.7799	0.0190
	VPint	45.017	0.9855	0.8455	0.8789	0.0092	47.056	0.9946	0.8474	0.8723	0.0075	45.137	0.9887	0.8097	0.8349	0.0083
	UTCR2	**49.590**	**0.9877**	**0.9451**	**0.9462**	**0.0045**	**53.384**	**0.9962**	**0.8853**	**0.9022**	**0.0037**	**51.519**	**0.9905**	**0.9215**	**0.9273**	**0.0040**

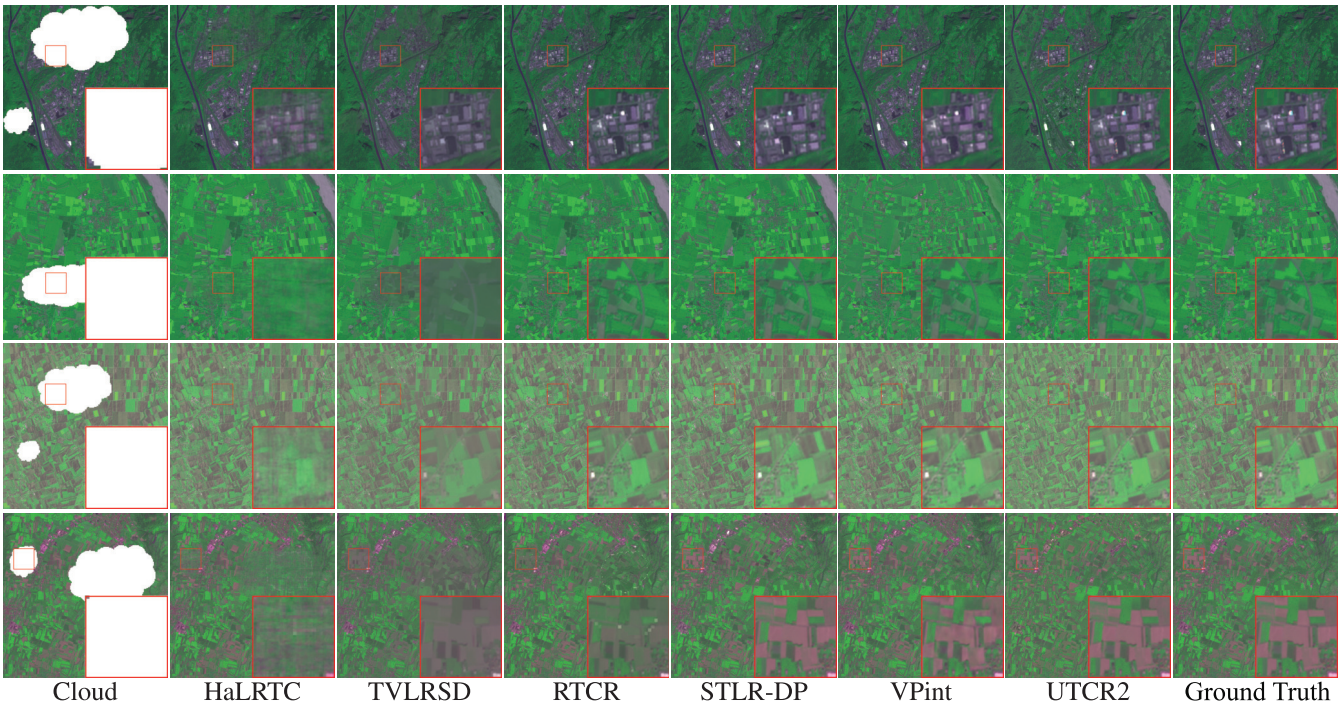


Fig. 5. False-color images composed of bands 3, 4, and 2 of the reconstruction results by different methods on the simulated experiments. (Top to bottom) The reconstruction results on *T32TPS* at time node 2023/08/14, *T32TQS* at time node 2023/08/16, *T33TUL* at time node 2024/03/05, and *T33TUM* at time node 2023/08/16, respectively.

properties of Sentinel-2 data, produces visually superior results with enhanced detail compared to competing methods.

To further analyze the effectiveness of all methods, we plot the scatter diagram in Fig. 6, which compares the reconstructed and original pixel values within cloud-affected areas across various scenarios. In Fig. 6, the x- and y-axis of the blue points denote the original and reconstructed pixel values, respectively. Generally, when the scatter points cluster closely around the diagonal line, it indicates superior performance of the methods in cloud removal tasks. Conversely, if the scatter points are widely dispersed and deviate significantly from the diagonal line, it suggests limited effectiveness of the cloud

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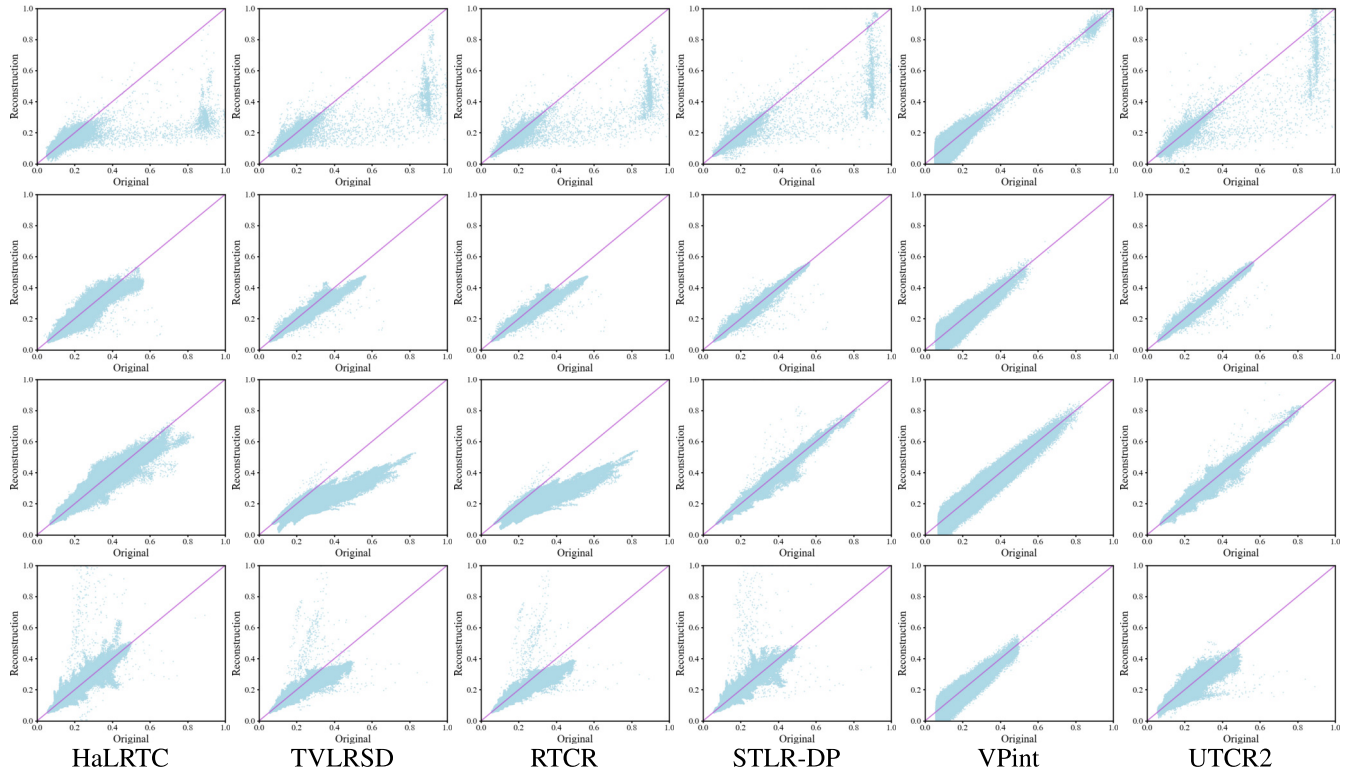


Fig. 6. Scatter plots comparing original and reconstructed pixel values in missing regions by different methods on simulated datasets. (Top to bottom) Scatter plots of *T32TPS*, *T32TQS*, *T33TUL*, and *T33TUM*. (Left to right) Scatter plots of the cloud removal results by HaLRTC, TVLRSD, RCTR, STLR-DP, and UTCR2, respectively.

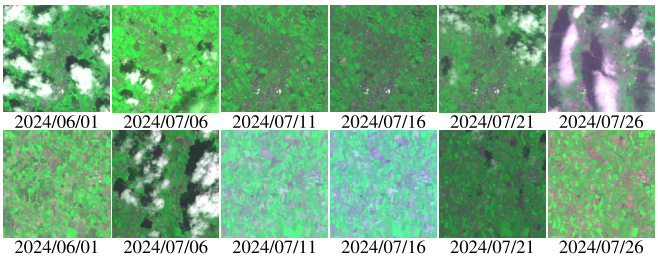


Fig. 7. Visual display of the false-color images composed of bands 3, 4, and 2 of (top) *T32TPR* and (bottom) *T32TQR* for real experiments.

TABLE II

AVERAGE NUMERICAL RESULTS OF THICK CLOUD REMOVAL BY DIFFERENT METHODS ON T32TPR AND T32TQR. THE BEST AND SECOND-BEST VALUES ARE HIGHLIGHTED

Dataset	*T32TPR*			*T32TQR*		
Methods	NIQE↓	BRISQUE↓	PIQE↓	NIQE↓	BRISQUE↓	PIQE↓
Observed	5.869	29.610	53.825	3.729	25.618	48.406
HaLRTC	5.152	26.381	<u>36.138</u>	3.826	25.884	39.788
TVLRSD	6.947	53.493	62.193	4.832	44.435	58.652
RTCR	4.251	26.001	44.254	4.232	25.866	<u>37.690</u>
STLR-DP	4.123	22.006	40.599	3.484	25.651	46.053
VPint	<u>3.889</u>	<u>20.774</u>	37.657	<u>3.343</u>	<u>22.632</u>	44.151
UTCR2	**3.665**	**14.023**	**24.623**	**3.255**	**21.233**	**36.700**

removal methods. From Fig. 6, it is clear that a large number of points of the reconstruction results by HaLRTC diverge from the diagonal lines, reflecting its inadequate reconstruction performance. This trend consists of the visual reconstruction results displayed in Fig. 5. Though RTCR achieves a tighter clustering of scatter points compared to HaLRTC, TVLRSD, and STLR-DP, some dispersion still remains. Conversely, the scatter points of the reconstruction results by the proposed UTCR2 are tightly aligned along the diagonal, indicating its superior performance and the best reconstruction results among all methods.

C. Real Experiments

For a comprehensive comparison, two real Sentinel-2 L2A/B datasets are selected. Because the original sizes of these datasets are too large, we crop subimages corrupted by clouds with spatial sizes of 600×600, 300×300, and 100×100 for the 10, 20, and 60 m SRs, respectively. For the competing methods, we use DSen2 [51] to rescale full bands of the real Sentinel-2 datasets into 10 m SR. Then, we perform all the competing methods on the real Sentinel-2 datasets rescaled by DSen2. Different from the competing methods, the proposed UTCR2 can directly use the original multitemporal Sentinel-2 image with different SRs as input without any preprocessing. The corresponding cloud masks are also acquired from the website.[2] If cloud masks are unavailable, the cloud mask can be estimated by many available cloud detection algorithms [52], [53]. The false-color images of *T32TPR* are shown in Fig. 7.

Since ground truth is unavailable in real data experiments, we adopt three widely recognized no-reference image quality assessment metrics: naturalness image quality evaluator

[2]https://browser.dataspace.copernicus.eu/

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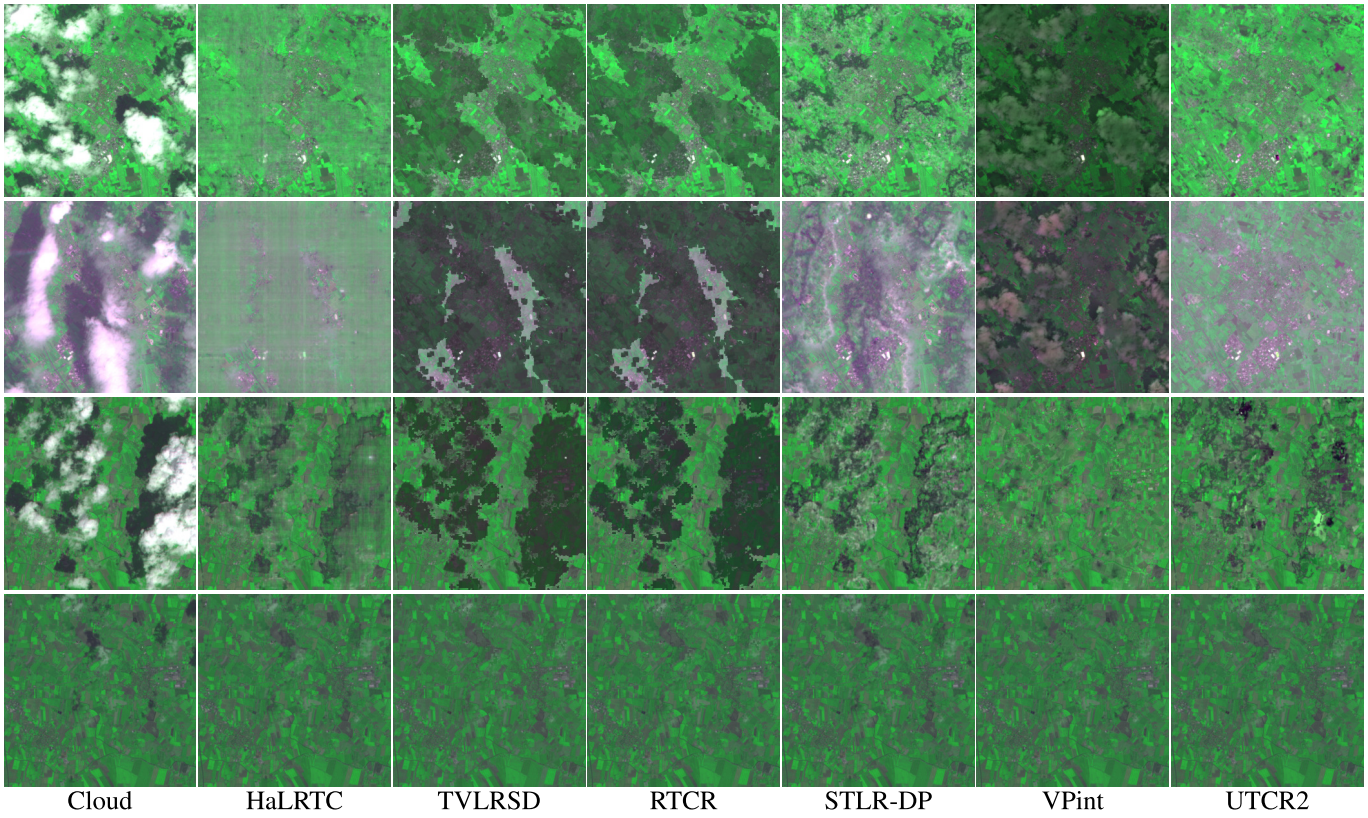


Fig. 8. False-color images composed of bands 3, 4, and 2 of the reconstruction results on *T32TPR* and *T32TQR* by different methods in real-world experiments. (Top to bottom) The reconstruction results on *T32TPR* at time node 2024/06/01, *T32TPR* at time node 2024/07/26, *T32TQR* at time node 2024/07/06, and *T32TQR* at time node 2024/07/21, respectively.

(NIQE) [54], blind/referenceless image spatial quality evaluator (BRISQUE) [55], and perceptual image quality evaluator (PIQE) [56] to assess the effectiveness of different methods. For these metrics, lower scores correspond to better perceptual quality, as they reflect reduced distortion and higher visual fidelity in the restored images. From Table II, we can observe that our proposed UTCR2 quantitatively outperforms all compared methods. The advantage of our proposed UTCR2 can be attributed to the fact that our proposed UTCR2 can genuinely leverage the full spectral bands of multitemporal Sentinel-2 images with different SRs.

Fig. 8 presents the false-color reconstructed results obtained by various methods applied to *T32TPR* and *T32TQR*. In general, the proposed UTCR2 demonstrates superior performance in restoration quality compared to the competing methods. As illustrated in Fig. 8, TVLRSD yields better visual reconstruction compared to HaLRTC. RTCR struggles to preserve fine details in its results due to neglecting spatial correlations in Sentinel-2 images. Similarly, the reconstructed results of STLR-DP are unsatisfactory in terms of spectral fidelity, particularly due to noticeable color distortion, which stems from the inconsistency between the distributions of the training and testing datasets. The proposed UTCR2 effectively incorporates the advantages of low-rank representation and INRs, leading to the visually promising results with enhanced detail preservation. The experimental findings highlight that UTCR2 exhibits robust stability and strong generalization capabilities in effectively removing thick clouds, further validating its superior performance.

TABLE III
NUMERICAL RESULTS OF THICK CLOUD REMOVAL BY THE PROPOSED UTCR2 WITH AND WITHOUT CIAM ON T32TPS. THE BEST VALUES ARE HIGHLIGHTED

Time Node	2023/08/14		2023/08/19		2023/08/26	
UTCR2	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
w/o CIAM	39.23	0.989	32.57	0.994	42.83	0.994
with CIAM	**42.69**	**0.993**	**34.50**	**0.995**	**45.68**	**0.996**

VI. DISCUSSIONS

A. Influence of CIAM

To thoroughly validate the influence of the CIAM integrated into the proposed UTCR2, an ablation study is conducted. Table III provides the PSNR and SSIM values of the reconstructed results by the proposed UTCR2 with and without CIAM (i.e., replacing the intrinsic abundance maps with learnable multidimensional arrays). As illustrated in Table III, it is evident that the carefully designed CIAM significantly enhances reconstruction performance. Furthermore, the proposed UTCR2 with CIAM surpasses UTCR2 without CIAM in terms of performance. These results underscore the critical role and effectiveness of the CIAM in improving the overall restoration quality. The improved performance can be attributed to the module's ability to model complex relationships within the data, ensuring that both fine details and broader structural features are preserved.

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TABLE IV

NUMERICAL RESULTS OF THICK CLOUD REMOVAL BY THE PROPOSED UTCR2 WITH AND WITHOUT DCSM ON T32TPS. THE BEST VALUES ARE HIGHLIGHTED

Time Node	2023/08/14		2023/08/19		2023/08/26	
UTCR2	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
w/o DCSM	40.23	0.990	33.57	0.994	43.53	0.994
with DCSM	**42.69**	**0.993**	**34.50**	**0.995**	**45.68**	**0.996**

TABLE V

NUMERICAL RESULTS OF THICK CLOUD REMOVAL BY THE PROPOSED UTCR2 ON T32TPS IN DIFFERENT CASES. THE BEST VALUES ARE HIGHLIGHTED

Time Node	2023/08/14		2023/08/19		2023/08/26	
SRs	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
10 m	45.39	0.986	35.58	0.989	50.88	0.994
10 m+20 m	49.69	0.990	40.85	0.992	54.08	0.998
10 m+20 m+60 m	**53.36**	**0.998**	**41.92**	**0.998**	**57.06**	**0.999**

B. Influence of DCSM

To thoroughly assess the influence of the DCSM, we conducted a comparative analysis of the performance of the proposed UTCR2 with and without DCSM (i.e., replacing the spectral signatures with learnable multidimensional arrays) on the *T32TPS*. Table IV provides a detailed comparison of the PSNR and SSIM values for the reconstructed results generated by UTCR2 with and without DCSM. As clearly demonstrated in Table IV, UTCR2 equipped with DCSM achieves notably higher PSNR and SSIM values compared to its counterpart lacking DCSM. This substantial improvement underscores the critical role of DCSM in enhancing the overall reconstruction performance, validating its effectiveness in refining the restoration quality of the proposed method.

C. Importance of Crossing Different SRs

To verify the importance of crossing different SRs, we compare the performance of the proposed UTCR2 on *T32TPS* with different cases. In 10 m case, the input of the proposed UTCR2 only includes multitemporal Sentinel-2 images with 10 m SR. In $10 + 20$ m, the input of the proposed UTCR2 only includes multitemporal Sentinel-2 images with 10 and 20 m SRs. In $10 + 20 + 60$ m case, the input of the proposed UTCR2 includes multitemporal Sentinel-2 images with 10, 20, and 60 m SRs. From Table V, we can observe that UTCR2 performed in $10 + 20 + 60$ m case significantly outperforms that performed in 10 m and $10 + 20$ m in terms of PSNR and SSIM values. Especially, UTCR2 achieves around 1 dB more than UTCR2* in terms of PSNR, which demonstrates that exploiting Sentinel-2 images across different SRs is beneficial for Sentinel-2 image thick cloud removal.

D. Importance of Multitemporal Resolution

To verify the importance of multitemporal resolution in Sentinel-2 image thick cloud removal, we compare the performance of the proposed UTCR2 on *T32TPS* with different numbers of time nodes, denoted as *T*. Table VI reports the PSNR and SSIM values of different methods. From Table VI, we observe that the PSNR and SSIM values rise as the number of time nodes increases. When all time nodes are utilized, the proposed UTCR2 achieves optimal performance. This observation indicates that simultaneously exploiting multitemporal resolution information is beneficial for Sentinel-2 image cloud removal.

TABLE VI

NUMERICAL RESULTS OF THICK CLOUD REMOVAL BY UTCR2 ON T32TPS WITH DIFFERENT NUMBERS OF TIME NODES. THE BEST VALUES ARE HIGHLIGHTED

SR	10 m		20 m		60 m	
Number of Time Nodes	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
1	35.296	0.9467	30.088	0.9698	35.712	0.9543
2	37.396	0.9567	31.588	0.9756	39.805	0.9613
3	40.669	0.9731	34.571	0.9828	44.772	0.9724
4	45.396	0.9867	35.758	0.9898	50.880	0.9815
5	49.985	0.9903	38.985	0.9910	53.056	0.9943
6	**53.364**	**0.9975**	**41.924**	**0.9977**	**57.065**	**0.9985**

TABLE VII

NUMERICAL RESULTS OF HYPERSPECTRAL IMAGE CLUSTERING ON CLOUD REMOVAL RESULTS BY DIFFERENT METHODS ON T32TPS. THE BEST VALUES ARE HIGHLIGHTED

Time Node	2023/08/14		2023/08/19		2023/08/26	
Methods	SC↑	DBI↓	SC↑	DBI↓	SC↑	DBI↓
HaLRTC	0.327	0.822	0.359	0.792	0.339	0.840
TVLRSD	0.347	0.800	0.359	0.786	0.358	0.783
RCTR	0.349	0.796	0.364	0.763	0.342	0.838
STLRDP	0.346	0.789	0.358	0.775	0.361	0.784
VPint	0.314	0.845	0.332	0.758	0.319	0.876
UTCR2	**0.358**	**0.783**	**0.367**	**0.754**	**0.373**	**0.749**

E. Downstream Task

To further evaluate the effectiveness of the proposed UTCR2, we considered a downstream hyperspectral clustering task on the *T32TPS* after thick cloud removal. We use the cloud-free image and the outputs of different cloud removal methods as the inputs to the clustering task, which is performed by a *K*-means classifier. To evaluate the performance of hyperspectral clustering, two different metrics are selected, including the silhouette coefficient (SC) and the Davies-Bouldin index (DBI). Generally, larger SC and smaller DBI values indicate better clustering performance. Table VII lists the numerical results of the hyperspectral image clustering. From Table VII, we can observe that the clustering on the cloud removal results by the proposed UTCR2 achieves the best performance in terms of SC, DBI, and CHI values. Fig. 9 displays the clustering results on the cloud removal results by different methods in 10 m SR. From Fig. 9, we can see that the cloud removal methods can improve the clustering performance by simply using the cloud-contaminated data. In addition, we can observe that clustering performance on the cloud removal result by UTCR2 is the closest to that on the cloud-free image.

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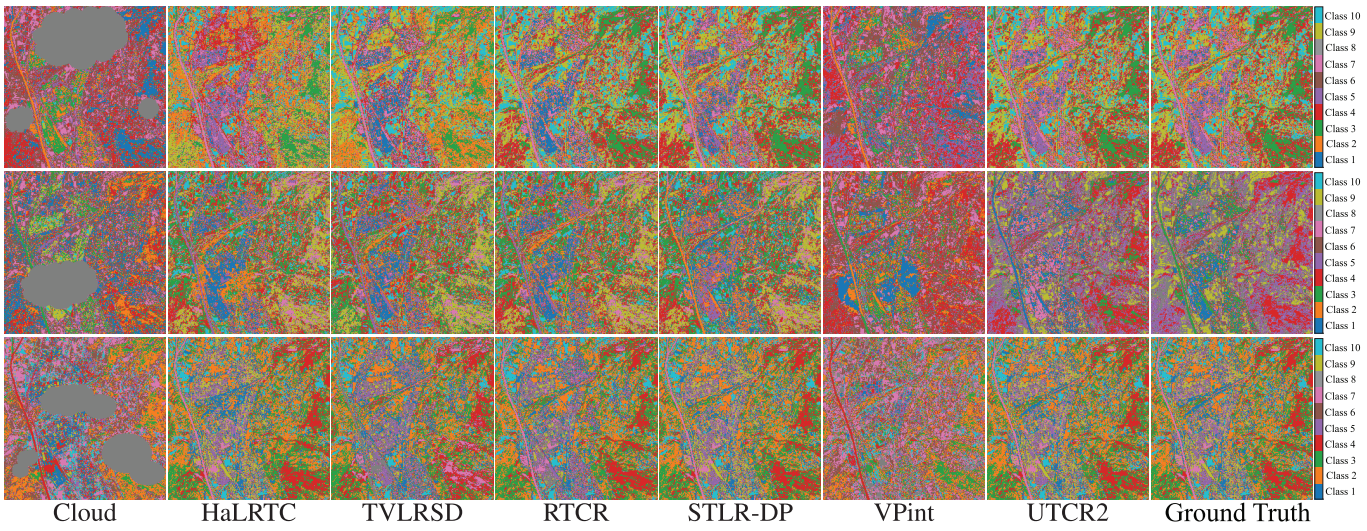


Fig. 9. Clustering results on the cloud removal results by different methods on *T32TPS*. (Top to bottom) The clustering result at time node 2023/08/14, 2023/08/19, and 2023/08/26, respectively.

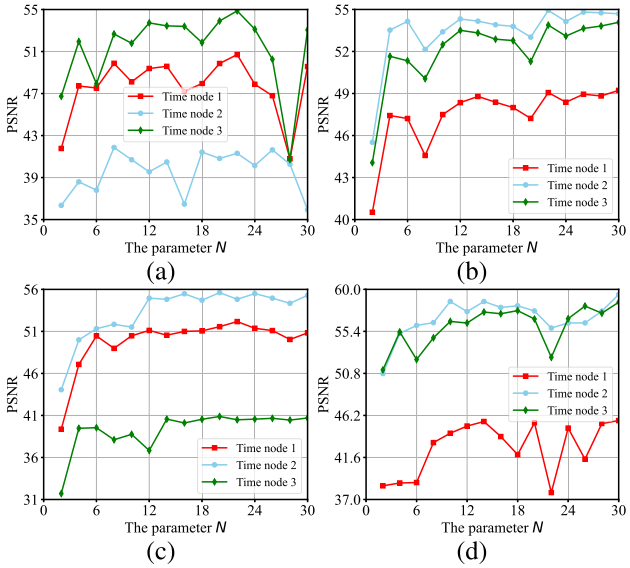


Fig. 10. PSNR values of the reconstruction results achieved by UTCR2 with varying N across various simulated datasets. (a) *T32TPS*. (b) *T32TQS*. (c) *T33TULI*. (d) *T33TUM*.

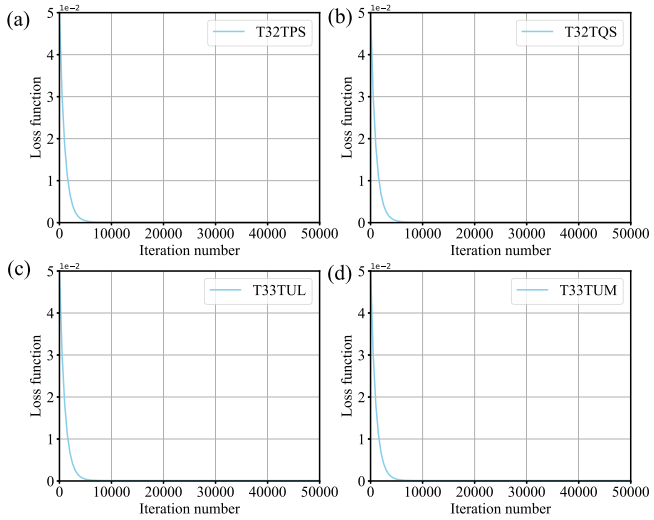


Fig. 11. History of loss function values across iteration counts for various simulated datasets. (a) *T32TPS*. (b) *T32TQS*. (c) *T33TUL*. (d) *T33TUM*.

F. Sensitivity of the Number of Endmembers N

We analyze the sensitivity of the number of endmembers N in this part. As shown in Fig. 10, the PSNR values of the reconstruction results obtained by UTCR2 vary with different N on simulated datasets. Specifically, PSNR increases as N grows within a certain range before reaching a plateau. Based on this observation, we recommend selecting N in the candidate set $\{12, 16, 18, 20, 24\}$. Beyond quantitative evaluation, existing endmember estimation algorithms [57], [58], [59] can also aid in determining an appropriate choice for N.

G. Convergence Analysis

To validate the convergence behavior of the proposed UTCR2, we plot the training loss curves versus he iteration count for various simulated datasets in Fig. 11. As illustrated in Fig. 11, it is evident that the training loss exhibits a consistent monotonic decrease and progressively approaches zero as the number of iterations increases. This trend is observed across all tested datasets, indicating stable and reliable convergence behavior. The empirical results presented here robustly confirm the convergence of the proposed UTCR2.

VII. Conclusion

In this paper, we suggest a unified continuous perspective for multitemporal Sentinel-2 image thick cloud removal and rescaling. The proposed method can continuously represent multitemporal Sentinel-2 images with different SRs. In the proposed method, we designed a CIAM to capture intrinsic abundance maps with varying SRs and a DCSM to capture the spectral signals of Sentinel-2 images. Extensive experiments on simulated and real data demonstrated the superiority of the proposed method over state-of-the-art thick cloud removal methods.

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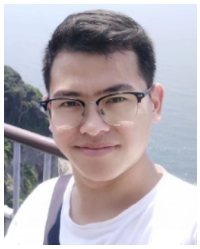
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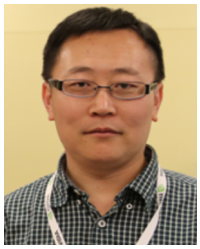
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